

A Geometrical Theory of Communication

Measurement, Meaning and Representation

Human argument with AI formalism and analysis

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Thesis, presentation and editorial direction

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Abstract

What must survive a change of description for information to retain its meaning? Shannon measures the uncertainty of selecting a message; the interpretation of its alternatives is supplied by the communication setting. This paper develops the complementary question of how that interpretation can be checked. Its thesis is that information, understood semantically, is invariance over change. The geometric bit operationalizes this thesis through a circular comparison: a reference, a perpendicular change, a symmetry and a test of return. Euler's formula expresses their relation. Following the author's visual presentation, the argument moves from ordinary language to binary geometry, measured identity, and composition. Numbered formal analyses derive reference-invariant comparison, discrete readings of the circle, the compatible Hopf hierarchy over the real normed division algebras, and exact transport of a single error through unit products. An aligned local difference distinguishes a missing nonidentity factor from a noncommuting elementary move; a trace version applies to unitary matrices. Meaning and action become operational tests when the reference, interpretation and expected outcome are specified. The larger identification of these structures across physics, computation and language is stated as the paper's research thesis. This edition separates that human argument from AI-developed formalism and records the assumptions and review status of each step.

Keywords: semantic information; identity; measurement; Euler's formula; geometric bit; Hopf fibration; normed division algebra; ordered composition; human–AI collaboration.

How this edition was made, and how to read it

This manuscript is being developed in public. Walter supplied the argument, examples, completed presentation and research direction. Astra prepared this reconstruction, formal analyses, figures and checks. Claude contributed the earlier analyses preserved in the source notes. Claude's review of this edition and Walter's approval of its wording are pending; their status is recorded in the accompanying review ledger.

The prose and figures carry the human argument, edited and expanded for a paper. The numbered boxes supply definitions, assumptions, proofs and their explanations, distinguishing established results from the thesis's interpretations. Each box has a stable identifier for discussion between the collaborators, through Walter, who makes the editorial decisions. Readers can follow the prose first and return to the formal development at each step.

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1 What is information?

What is information, and what does it mean to know something? We use the word easily: news, data, a measurement, something we did not know before. Each answer points toward a change in what is available to us, but it also assumes something that the information is about. Uncertainty has an object; there is a question whose answer we have yet to learn.

Let us say that information exists, that I have observed it and measured it, and that I know what it is. How do I give you that knowledge? A paper about information is already inside its own subject, because everything written here has to pass through signs that you read and understand. The same holds for every science: descriptions carry observations beyond the person who made them, and another person must be able to return from the description to something they can check. Hertz made this relation between our representations and their consequences central to his account of mechanics (Hertz, 1899).

Say I point to a lighter on the desk between us. You see it from one side and I see it from another, yet we can both reach for the thing I meant. I could call it an igniter and change the word without changing what I am pointing at. A photograph, a spoken name and a measurement of its weight give different descriptions, each making some relationship available to someone else.

Looking already has three parts: the observer, the observed and the observation. You, the lighter and the light that lets you see it belong to one encounter. A weighing has the same structure, with the scale and its calibration making the comparison explicit. Peirce develops a related triadic account of signification through sign, object and interpretant, the understanding or effect the sign produces (Peirce, 1904). These vocabularies attend to different parts of the encounter, while both keep the mediating relationship in view.

This gives the paper its three questions: *What exists? What do we measure/observe? How do we convey it?* Physics enters through the interaction, epistemology through the warrant for a reading, and language through the description another person receives. Mathematics makes the relationships precise enough that an operation performed on a representation can be checked against the operation it represents.

AI formalism and analysis F01: An observation and its representation

Status: model definition.

Let X be a set of possible systems or states, R a set of measurement contexts, and Y a set of readings. An observation has the form

$$M : X \times R \longrightarrow Y, \quad y = M(x, r). \quad (1)$$

The context r can specify an instrument, units, orientation, time and calibration. An encoding $E : Y \times R \rightarrow \mathcal{S}$ then produces a sign $s = E(y, r)$ in a signal alphabet $\mathcal{S}$.

The same inscription can result from different pairs (x, r) . To interpret it, a receiver needs whatever part of r affects the question being answered. For example, the numeral 90 acquires a rotational meaning through its angular unit: degrees and radians prescribe different operations.

Equation (1) formalizes the three roles in a comparison. It imposes no claim that all physical interactions require three independent particles or three spatial dimensions. A stochastic instrument is represented by a conditional distribution $p(y | x, r)$ in place of a deterministic M .

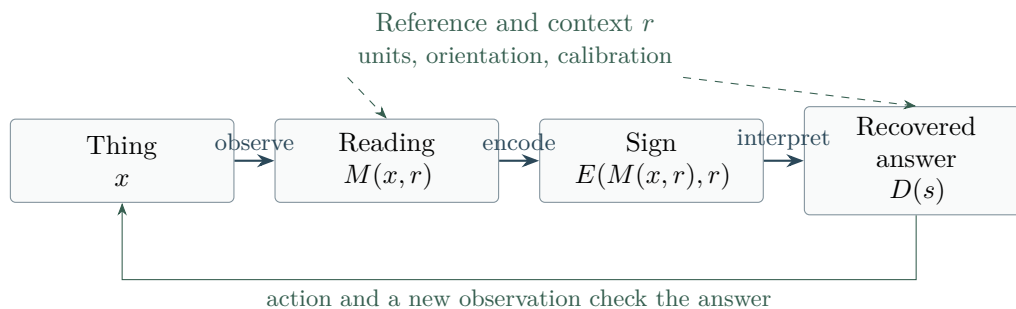


Figure 1: The paper follows a relationship through observation, representation and response. Calibration makes the observation interpretable; a reference and a check make the receiver’s response comparable with what was intended.

1.1 A note on interdisciplinarity

I came to the question from psychology: how can two people know that they have understood one another? Asking for a definition produces another description; asking them to do something with the description gives us an outcome to compare. Following that comparison reaches measurement, and following measurement reaches the relationships that make a reading repeatable.

The disciplines therefore enter through a common problem. A physical record must be realized somewhere, as Landauer emphasizes (Landauer, 1996); a mathematical description must specify its objects and operations; a psychological account must explain what a person can do with a relation they have learned. Keeping those obligations together is the method of this paper. The circle will provide a case simple enough to follow through every step, before we ask how its comparison extends to richer objects.

2 A communication problem

As soon as I know something and set out to tell you, we have a communication problem. The word *bank* can reach you perfectly while leaving open whether I mean a financial institution or the edge of a river. The signal arrived; understanding it calls on the setting in which it was used. Here we are also using language to discuss language, so the question returns: how do I convey information about information?

Weaver separates three questions in his introduction to Shannon’s theory (Weaver, 1949). Level A concerns the accuracy of transmission. Level B concerns whether the symbols convey the intended meaning. Level C concerns whether that meaning leads to the intended effect. A person can repeat a sentence without understanding its use, and can understand a request while choosing a different action. Each level needs its own observation.

The present inquiry concerns Level B and the way it can be checked through Level C. In this semantic use, information is what remains available across different expressions: the lighter reached through *lighter* and *igniter*, or one rotation described in degrees and radians. Communication requires enough agreement that what I mean can become what you understand. Mathematical language makes that agreement exact by specifying what its symbols denote and what operations on them do.

Frege’s contrast between $a = a$ and $a = b$ helps explain why sameness can teach us something (Frege, 1892). Learning that the morning star and the evening star are Venus adds a route between descriptions of the same object. The object does not have to change for the reader’s knowledge to change. A successful message connects a new description to something that can

be recognized.

AI formalism and analysis F02: The semantic round trip

Status: definition and exact recovery criterion.

Let A be the set of meanings relevant to a specified task, S a set of messages, and $E : A \rightarrow S$, $D : S \rightarrow A$ an encoder and decoder with a shared interpretation. Exact semantic recovery is

$$D(E(a)) = a \quad \text{for every } a \in A. \quad (2)$$

Consequently E is injective: if $E(a) = E(b)$, applying D gives $a = b$. Different signs may still decode to one meaning, because D need not be injective. Thus *lighter* and *igniter* may have the same decoded value.

When a context r determines interpretation, use E_r and D_r . A technical channel C may satisfy $C(s) = s$ while a receiver uses $D_{r'}$ with $r' \neq r$; then $D_{r'}(C(E_r(a)))$ can differ from a . This is the mathematical shape of the bank example. The condition states what successful communication means after the task and interpretations have been specified.

If the instruction is to rotate a pointer through a quarter cycle, one person can write 90° , another $\pi/2$, and another multiplication by i . Reading the numeral 90 as radians sends the pointer elsewhere even if every character crossed the channel intact. A return test compares the resulting rotation with the intended one. This is where the identity principle becomes an operation: we carry out a transformation and check what it preserved.

AI formalism and analysis F03: Preserving an operation as well as a name

Status: representation criterion.

Let $T : X \rightarrow X$ be an allowed transformation and $E : X \rightarrow Z$ a representation. Let $\hat{T} : Z \rightarrow Z$ denote the represented operation. The required agreement is

$$E(Tx) = \hat{T}(E(x)). \quad (3)$$

This is equivariance: performing the operation before or after encoding gives the same represented result. For degrees and radians, $E(\theta) = \pi\theta/180$ converts the units, and adding 90 degrees becomes adding $\pi/2$ radians.

Together, (2) and (3) say what a usable translation must retain: the relevant answer and the operations that make it useful. They also provide a way to test a learned representation without judging it only by resemblance between its symbols.

3 Shannon's bit

How did Shannon make the communication problem manageable? He began with the messages that could be selected and asked how reliably their selection could be reproduced elsewhere. Meaning was set aside because the engineering system had to work for whatever interpretation the users assigned to those messages (Shannon, 1948). Weaver explains the measure as belonging to the situation of choice, rather than to the significance of a particular utterance (Weaver, 1949).

His example makes the distinction easy to see. Suppose the two possible messages are the whole King James Bible and the single word “Yes”. If both ends already possess the alternatives and their code, one signal selects either of them. The receiver can use 0 for the first and 1 for the second. The choice can be small even when the selected object is large, because the description needed to identify that object has already been shared.

The bit counts this choice between two equally likely alternatives as one. Adding another independent binary choice doubles the possibilities, and two bits distinguish four outcomes. Three distinguish eight. The logarithm enters because the number of combinations multiplies while the number of choices adds. Unequal probabilities then refine the count: an outcome we expected with certainty resolves nothing, while an unlikely outcome resolves more.

AI formalism and analysis F04: Choice, probability and repetition

Status: established information theory.

For a discrete random message X with probabilities $p(x)$, define

$$I(x) = -\log_2 p(x), \quad H(X) = -\sum_x p(x) \log_2 p(x). \quad (4)$$

Here $I(x)$ is the information associated with an outcome and $H(X)$ its expected value. For N equally likely messages, $H(X) = \log_2 N$. Independent choices satisfy $H(X, Y) = H(X) + H(Y)$; in general,

$$H(X_1, \dots, X_n) = \sum_{k=1}^n H(X_k | X_1, \dots, X_{k-1}). \quad (5)$$

Shannon develops these properties and their coding interpretation (Shannon, 1948).

A source known to send only 0 has $H = 0$, and the same is true of a source known to send only 1. If the source first chooses fairly between 000000 and 111111, the whole message contains one bit, despite its repetition. Independent fair symbols contain one bit each, including a realized run of zeros. These examples make the relationship precise: information depends on which possibilities could have occurred and which remain after observation. The appearance of a mixed string alone does not determine its entropy.

Binary gains its reach by combining symbols with position. In 1101, the positions carry weights 8, 4, 2, 1, so the numeral represents thirteen. Numbers can then index letters, colour levels or sound samples. Each use has a reference: an alphabet, a pixel arrangement, a clock, a unit. What can be sent depends on the distinctions those references make available.

Every physical bit also occupies a physical state, such as two voltage ranges or two magnetization directions. To draw the distinction geometrically, put two marks one unit apart. With two bits, let each vary while the other remains fixed: the four possibilities become the corners of a square. Three bits make a cube, and further bits give its higher-dimensional versions. Boolean analysis already works directly on this geometry (O'Donnell, 2014).

AI formalism and analysis F05: The binary cube and its operations

Status: established construction.

For n labelled positions the state set is $\{0, 1\}^n \subset \mathbb{R}^n$, with the usual orthonormal coordinate basis. Two strings x, y have Hamming distance

$$d_H(x, y) = \sum_{k=1}^n |x_k - y_k| = \|x - y\|_2^2. \quad (6)$$

Each coordinate records a separately addressable distinction. For two inputs, $\text{AND}(x, y) = xy$ and $\text{XOR}(x, y) = x + y - 2xy$ agree with their truth tables at all four corners. A code corrects up to t bit flips when its minimum distance is at least $2t + 1$ (Hamming, 1950).

This is a geometric representation with a chosen metric. Coordinate orthogonality expresses separation of readings; probabilistic independence is an additional property of a distribution. The ordered strings 01 and 10 remain different corners. Binary coding can therefore preserve order when its positions preserve it.

Noise changes what arrives. A scratch may flip a bit, and a weak signal may leave two alternatives hard to distinguish. Shannon asks how much information can pass reliably through a specified channel, allowing long codes to protect the choices. This question remains part of the present theory: giving a meaning a geometric representation still leaves us with a physical channel through which that representation must travel.

AI formalism and analysis F06: Capacity and the continuous channel

Status: established results with their models.

For a discrete memoryless channel $p(y | x)$, capacity per channel use is

$$C = \max_{p(x)} I(X; Y), \quad I(X; Y) = H(X) - H(X | Y). \quad (7)$$

Rates below capacity admit arbitrarily small decoding error with sufficiently long suitable codes. This statement depends on the channel model and coding assumptions. For an ideal real band-limited additive white Gaussian noise channel of bandwidth B , average signal power P and noise power N in the band, the corresponding rate is

$$C = B \log_2(1 + P/N) \quad \text{bits per second.} \quad (8)$$

Gaussian noise is this particular continuous example, not an assumption of all Shannon theory (Shannon, 1948, 1949).

For a signal whose Fourier transform vanishes outside $[-B, B]$, suitable sampling at interval $1/(2B)$ reconstructs it through

$$f(t) = \sum_{k \in \mathbb{Z}} f\left(\frac{k}{2B}\right) \text{sinc}(2Bt - k), \quad \text{sinc}(u) = \frac{\sin(\pi u)}{\pi u}, \quad (9)$$

under the usual bandlimit and convergence hypotheses. The samples retain phase and structure when the model and clock are known. A time window T has roughly $2BT$ effective real degrees of freedom; exact simultaneous compact time support and band support are different requirements.

The first part of the story is now in place. A bit counts alternatives, entropy measures uncertainty over them, and a channel limits reliable transmission. The alternatives themselves must already be identifiable. The question for meaning is how to carry and check the relationships that let a receiver identify them after the description changes.

4 The geometric bit

What is the shape of a comparison that keeps its reference? Draw a circle and mark a point on it. Draw the diameter through that point and a second diameter at a right angle. The construction gives four positions before we name them: the starting point, its opposite, and the two positions halfway between. Writing $1, i, -1, -i$ records the quarter rotations that relate them.

Perpendicularity distinguishes two directions while preserving the unit used to compare them. Symmetry relates opposite positions through the same half-rotation. Follow the marked point around the boundary and its position changes continuously, while the radius and the relationships between the axes remain fixed. After a full rotation the point reaches its start. We can now show sameness through a change, carry out a comparison, and measure a failure to return.

This is the paper's answer to the question with which we began: *information is invariance over change*. The geometric bit gives that answer a form we can operate. It is the circular

relationship together with a reference and a comparison, so that identity is available as a check. The principle $A = A$ becomes something we can perform through a representation, transform and return, rather than leave implicit in the name of an alternative.

4.1 The alphabet of the circle

The alphabet of binary is 0 and 1, and DNA uses four bases in sequences whose organization matters to their function. Here the alphabet is relational: perpendicularity and symmetry. They describe how one direction differs from another and how the same relation appears again. The symbols enter after the construction, as a concise way of carrying it.

Why does i appear? A quarter rotation applied twice reaches the opposite; applied four times it returns. On the plane, the operation sending a horizontal unit vector to a vertical one therefore has a square equal to negation. Writing that operation as i gives $i^2 = -1$. The expression is a record of what the operation does.

AI formalism and analysis F07: From the right angle to Euler's formula

Status: elementary derivation.

On an oriented Euclidean plane choose an orthonormal basis and define

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^2 = -I, \quad J^\top = -J. \quad (10)$$

For a unit initial vector v_0 , the differential equation $v'(\theta) = Jv(\theta)$ has solution

$$v(\theta) = e^{\theta J} v_0 = (I \cos \theta + J \sin \theta) v_0. \quad (11)$$

The last equality follows by grouping the even and odd powers in the exponential series. Moreover, $\frac{d}{d\theta} \|v\|^2 = 2v^\top Jv = 0$, so the solution remains on the unit circle. Identifying J with multiplication by i gives

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad e^{i\pi} = -1, \quad e^{2\pi i} = 1. \quad (12)$$

The exponential accumulates the generator; the generator i makes that change perpendicular. The half-period π reaches the opposite. Thus the presentation's descriptions of e as accumulation and π as symmetry have precise roles in one equation. Real exponential growth and perpendicular rotation use different generators of the same exponential construction.

4.2 A unit that can be checked

The circle brings together a continuous path and a countable event. Its boundary can be followed continuously, while a completed rotation can be counted as one. When we cut a pizza into four equal quarters, the right angles belong to that construction before we assign numbers to the pieces. Choosing the first cut sets the orientation; making its perpendicular sets the other axis. Rotating the whole arrangement changes neither the quarters nor their relations.

The useful distinction is between the relation and its reading. We may record only which half contains the mark, record a quadrant, or measure an angle to a chosen precision. Each answer is a different description of the same comparison space. A finite alphabet becomes available by deciding which differences the reading must preserve.

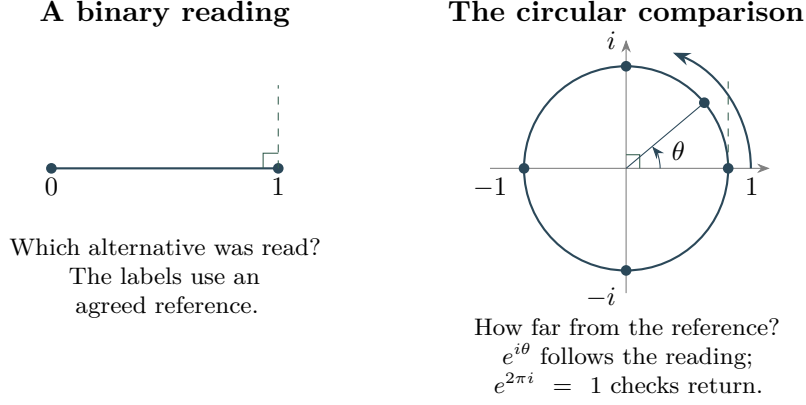


Figure 2: A discrete reading retains an alternative; the circular comparison also retains the reference and the operation relating readings. The two axes are perpendicular, the opposite poles are symmetric, and rotation preserves the circle. The radius and tangent at a point are perpendicular as well.

AI formalism and analysis F08: The geometric bit and its discrete readings

Status: definition and interpretation.

Define the circular comparison unit as

$$\mathcal{G} = (S^1, 1, \mu, J, d), \quad S^1 = \{z \in \mathbb{C} : |z| = 1\}, \quad \mu(z, w) = zw, \quad (13)$$

where 1 is the reference, $Jz = iz$ is the quarter rotation, and

$$d(z, w) = |\text{Arg}(\bar{z}w)| \in [0, \pi]. \quad (14)$$

This tuple is the *geometric bit* in this manuscript. Its identity check is $\bar{z}w = 1$, equivalently $d(z, w) = 0$. Its intended semantic role is to give the identity principle an operational representation. Choose $\theta \in [0, 2\pi)$ and define $Q_m(e^{i\theta}) = \lfloor m\theta/(2\pi) \rfloor$. For $m = 2$ this records a half-circle; for $m = 4$ it records a quadrant. A distribution of readings has Shannon entropy $H(Q_m(Z))$. This construction recovers binary alternatives as a coarse reading. The circle itself is a continuum, and “geometric bit” names the comparison structure rather than a claim that an arbitrary angle costs one Shannon bit to transmit.

Two people facing different directions can compare angles if each keeps the reference in the same description as the reading. Rotate the reference and the reading together, and their separation remains unchanged. This is the smallest complete example of what the lighter asks of us: our views can differ while a relation survives the change between them.

AI formalism and analysis F09: The reference travels with the reading

Status: exact invariance.

For reference $r \in S^1$ and reading $z \in S^1$, let $c(r, z) = \bar{r}z$. Under a common frame change $u \in S^1$,

$$c(ur, uz) = \overline{ur}uz = \bar{r}\bar{u}uz = \bar{r}z = c(r, z). \quad (15)$$

Two observers therefore recover the same relative phase from differently oriented coordinates. If they transmit only z and omit an unknown r , the value of $c(r, z)$ is generally undetermined. A representation carrying both a reading and a usable reference solves this particular ambiguity. For an image or a physical object, the analogous construction needs an observation model. Sampling reconstructs the band-limited field of Box F06 from sufficient calibrated samples; recovering a three-dimensional lighter also involves viewpoint, illumination and occlusion. The shared-frame

AI formalism and analysis F09 (continued): The reference travels with the reading

calculation is exact for its stated circular task and supplies the pattern that a richer representation must implement.

4.3 Closure, counting and uncertainty

Following a point for one rotation or for two ends at the same place. We have to keep a count if the distinction between those journeys matters. The circle then gives two records: where the point is now and how many completed rotations brought it there. This is the continuous-to-discrete relationship behind the counting language of the presentation.

Closure also supplies a second question alongside uncertainty. Before the message arrives, which alternatives remain possible? After interpreting it, how closely does the result agree with the reference? Entropy answers the first, and a closure residual answers the second. The thesis treats them as complementary roles in knowing: a difference is measured against something that holds, and what holds is recognized through differences.

AI formalism and analysis F10: Endpoint, winding and closure residual

Status: established topology and defined measurement.

The covering map

$$p : \mathbb{R} \rightarrow S^1, \quad p(t) = e^{2\pi it}, \quad p^{-1}(1) = \mathbb{Z} \quad (16)$$

retains phase while identifying histories that differ by a whole number of rotations. A continuous path has a unique real lift once its initial lift is chosen. A closed path then has integer winding $n = \theta(T) - \theta(0)$ in turns, expressing $\pi_1(S^1) \cong \mathbb{Z}$ (Licata and Shulman, 2013).

For expected endpoint r and reconstructed endpoint z , define the closure residual $\varepsilon = d(r, z)$. It is zero exactly when the endpoints agree. If the task includes winding, compare (z, n) rather than z alone.

Entropy $H(X)$ is measured in bits and depends on a probability distribution; ε is measured in radians and depends on a reference. No universal constant makes their sum fixed. For example, a fair binary message can have entropy one and be recovered with zero residual on every trial. Closure is therefore a verification measurement alongside uncertainty, with its own specified object and unit.

5 Identity, measured

What changes once identity is measured instead of assumed? We can ask whether the relationship in the description is also the relationship an instrument finds. A string around a circular rim compares circumference with diameter. A clock compares equal intervals in a decay or oscillation. A plumb line and a locally level surface make a right angle available to construction. These are different ways of approaching the constants and operations in Euler's formula, through things we can do in the world.

The notation is ours to choose. We could name a quarter rotation in degrees, radians or turns, while the pointer reaches the same place. What the notation preserves is constrained by the measurement. This is the sense in which the paper calls Euler's identity a physical law of comparison: its circular relationship is realized and repeatedly checked by physical operations.

This claim has a mathematical layer and an empirical layer. The exact return follows in the rotation model; experiments establish how well an apparatus realizes that model. Holding the layers together gives the law practical content, because a discrepancy can be assigned to a unit,

a reference, a transformation or an instrument instead of being hidden inside an inscription.

AI formalism and analysis F11: The proposed law of comparison

Status: physical interpretation with an explicit model.

An ideal circular apparatus realizes a continuous, orientation-preserving rotation $R(\theta)$ with $R(0) = I$, a fixed Euclidean norm, and a calibrated angle satisfying $R(\alpha + \beta) = R(\alpha)R(\beta)$. With one revolution calibrated as 2π , Box F07 gives $R(2\pi) = I$ exactly. An experimental return has residual

$$\varepsilon_{\text{exp}} = d(z_{\text{initial}}, z_{\text{final}}), \quad (17)$$

reported with uncertainty from resolution, drift and calibration. Reproducing this relation across instruments is the empirical content of the proposed law. Its exact mathematical content belongs to the stated rotation model.

The circumference test assumes Euclidean planar geometry. On a sphere of radius a , a geodesic circle of radius s has circumference $2\pi a \sin(s/a)$, giving the ratio $\pi \sin(s/a)/(s/a)$ to $2s$. Local flatness recovers π as $s/a \rightarrow 0$. The phase identity remains exact while the metric interpretation of a particular circumference changes. A measurement must specify which circle and which distance it compares.

5.1 Why the same equation keeps appearing

Let change be proportional to what is already present and the exponential appears. Let that change point along the present state and it grows or decays; let it point perpendicularly and it rotates. A charged capacitor and a rotating phase therefore use the same accumulation rule with different generators. The numerical value of e belongs to the normalized exponential, while the physical rate tells us how laboratory time enters its exponent.

In mechanics, the oscillator moves around its energy contours. In quantum mechanics, unitary evolution keeps the length of the state vector. In signal processing, sines and cosines let a field be resolved into components whose amplitude and phase can be measured. These are substantial instances of the same geometric language. Each obtains its physical prediction from a generator and a model, as well as from the exponential that composes their effects.

AI formalism and analysis F12: Three concrete realizations

Status: established dynamics.

For $\dot{x} = kx$, $x(t) = x(0)e^{kt}$, where k has units of inverse time. For the harmonic oscillator, take normalized coordinates $Q = \sqrt{m\omega} q$ and $P = p/\sqrt{m\omega}$, and let $z = Q + iP$. Hamilton's equations give

$$\dot{z} = -i\omega z, \quad z(t) = e^{-i\omega t} z(0). \quad (18)$$

The energy is proportional to $|z|^2$, so it remains fixed during rotation. More generally, canonical Hamiltonian evolution has $\dot{x} = J\nabla H$, with antisymmetric J , and $\dot{H} = (\nabla H)^\top J \nabla H = 0$ for a time-independent Hamiltonian.

For a finite-dimensional closed quantum system with self-adjoint Hamiltonian $\hat{H}$, Schrödinger evolution is

$$i\hbar\dot{\psi} = \hat{H}\psi, \quad \psi(t) = e^{-it\hat{H}/\hbar}\psi(0). \quad (19)$$

It preserves $\langle\psi, \psi\rangle$ because the generator is skew-Hermitian. The velocity is perpendicular to the state in the real inner product $\text{Re}\langle\cdot, \cdot\rangle$; it need not be orthogonal in the complex inner product. Several incommensurate eigenfrequencies need not return together after a finite time. These statements explain precisely which rotation properties the physical examples share.

The same distinction helps with conservation. Noether connects continuous symmetries of

an action with conserved quantities under definite variational conditions (Noether, 1918). The circle offers a simple symmetry we can operate, while a conservation law also needs the dynamics to which that symmetry applies. Similarly, entropy needs a statistical model of alternatives; perpendicular axes provide a representation in which separate variations can be described. The thesis brings these roles into one comparison rather than treating the words as interchangeable equations.

5.2 What counts as a check?

To exist physically, in the operational account used here, is to be available to interaction. The lighter reflects light, exchanges heat, rests on the desk and resists a hand. Those relationships continue when we look away. The observer, the instrument and the inscription participate in the same world, so the description can be brought back to a comparison outside its own wording.

That is the epistemological use of consistency in this argument. Agreement between independent procedures strengthens the case that they preserve a common relationship. A string, a clock and an optical phase measurement can constrain the same model from different directions. Their agreement motivates the unifying interpretation; each observation still has a domain and a precision that can be examined by another person.

AI formalism and analysis F13: Operational identity and the scope of verification

Status: methodological definition.

Fix a family of admissible observations $\mathcal{M}$. Define

$$x \sim_{\mathcal{M}} y \iff M(x) = M(y) \text{ for all } M \in \mathcal{M}. \quad (20)$$

For stochastic observations, require equality of their outcome distributions. This is an equivalence relation, and the quotient $X/\sim_{\mathcal{M}}$ records precisely the distinctions those observations can resolve. Enlarging $\mathcal{M}$ can separate previously equivalent states.

An invariant $f(Tx) = f(x)$ preserves identity relative to allowed transformations T ; a *complete* invariant additionally distinguishes the equivalence classes relevant to the task. A circle's constant radius is an invariant, but radius alone does not identify an arbitrary physical object.

Formal verification and empirical checking address different parts of the chain. Gödel's incompleteness theorems concern consistent, effectively axiomatized systems of sufficient arithmetic strength (Gödel, 1931). External measurements give scientific models new evidence; they do not remove those hypotheses or convert every true statement into a formal proof. Here the evidence, the formal implication, and the interpretation are each made available for inspection.

The thesis is now explicit: the identity that supports communication is a relationship preserved through interaction and transformation, and geometry gives us an apparatus for measuring it. The circular construction establishes this directly for rotational comparisons. To understand what richer objects require, we next compose those comparisons and keep track of what survives.

6 Operating geometric bits

What can geometric bits do, and where do they stop? Shannon's theory gives a unit, a measure, rules for composing messages and limits on a channel. With the reference retained, the same questions become questions about a comparison that can be composed and checked. We need to know which structure composition preserves, how much a description keeps, and what a

discrepancy tells us.

Question	Shannon's quantitative account	Geometric comparison developed here
Unit	A bit of uncertainty	A referenced circular comparison
Measure	Entropy and mutual information	Relative phase and closure residual, along-side entropy
Composition	Ordered strings and joint signal spaces	Joint states and ordered algebraic products
Limit	Capacity of a specified channel	Norm-preserving algebraic composition under Hurwitz's hypotheses
Noise	Error probabilities and recoverable rate	Exact single-factor transport and aligned local discrepancies
Meaning	Interpretation supplied by the task	Interpretation and its verification included in the protocol

Table 1: The theories answer connected questions at different points of the communication chain. A geometric representation continues to incur coding, calibration and channel costs.

6.1 Side by side, and in sequence

There are two operations to distinguish at the outset. Put two rotations side by side and we can describe both at once. Let one operation act after another and their order can change the result. A photograph uses many sampled values side by side; rotating a camera successively about two different axes is an ordered composition. Both carry information, and each preserves different relationships.

Holding a list of complex amplitudes to one total norm gives a sphere. Two complex amplitudes supply four real coordinates with one norm constraint, so their unit states lie on the three-sphere. This is also the unit sphere of quaternions, which supplies an additional operation: multiplying one unit quaternion by another. The shared sphere lets the two descriptions meet; the multiplication tells us how to operate its points.

AI formalism and analysis F14: Three composition spaces

Status: established constructions.

For m phases with each amplitude fixed, the joint space is the torus $(S^1)^m$. Allowing the amplitudes to vary while fixing their total squared norm instead gives

$$\left\{ (z_1, \dots, z_m) \in \mathbb{C}^m : \sum_{j=1}^m |z_j|^2 = 1 \right\} = S^{2m-1}. \quad (21)$$

Here $z_j = r_j e^{i\theta_j}$, with $\sum r_j^2 = 1$. The phase of a zero component has no effect. A common phase acts by $z \mapsto e^{i\alpha} z$; its quotient is complex projective space $\mathbb{CP}^{m-1}$.

For $m = 2$, identify the same four real coordinates with $q = w + xi + yj + zk \in \mathbb{H}$, $|q| = 1$. Ordered composition uses Hamilton's rule

$$(a, \mathbf{u})(b, \mathbf{v}) = (ab - \mathbf{u} \cdot \mathbf{v}, a\mathbf{v} + b\mathbf{u} + \mathbf{u} \times \mathbf{v}). \quad (22)$$

This operation keeps the norm and retains order. The dot and cross products are the scalar and vector contributions of one multiplication. A normalized list, a torus of fixed phases and an ordered product are thus three explicit constructions, rather than three names for one unspecified composition.

To see what order adds, rotate a physical frame about one axis and then another. Inter-

changing the rotations changes where the frame points. Complex phases compose by adding angles and therefore commute, while quaternionic rotations record the difference between the two orders. Perpendicularity now enters the multiplication itself, through the cross product.

6.2 The algebraic limit and the nested fibrations

How far can this construction grow while multiplying lengths exactly? The question has a sharp mathematical answer once its requirements are stated. Over the reals, the finite-dimensional unital algebras with a positive multiplicative norm occur in dimensions one, two, four and eight. They are the real numbers, complex numbers, quaternions and octonions. This is Hurwitz's classification, presented with its geometric consequences by [Baez \(2002\)](#).

The succession matters because each level can carry the previous one. A chosen complex plane still rotates by Euler's formula inside the quaternions or octonions. New independent directions enrich composition while preserving that local comparison. At the same time, the operation changes: quaternions give up commutativity, octonions give up associativity, and the next Cayley–Dickson doubling loses the multiplicative norm.

AI formalism and analysis F15: The limit theorem used in this paper

Status: established classification.

Let A be a finite-dimensional real algebra with identity, bilinear multiplication and a positive-definite quadratic norm satisfying

$$|ab| = |a| |b| \quad \text{for every } a, b \in A. \quad (23)$$

Then A is isomorphic to $\mathbb{R}, \mathbb{C}, \mathbb{H}$ or $\mathbb{O}$, of real dimensions 1, 2, 4, 8. The octonions are alternative: every two elements generate an associative subalgebra. The theorem concerns this specified algebraic class. Arbitrarily large vector spaces, matrix algebras, tensor products and composite quantum systems remain available outside it.

Norm preservation also preserves perpendicularity. For a unit a , the map $L_a : x \mapsto ax$ is real-linear and preserves lengths. By polarization,

$$\langle ax, ay \rangle = \frac{1}{2} (|a(x+y)|^2 - |ax|^2 - |ay|^2) = \langle x, y \rangle. \quad (24)$$

The same holds for right multiplication. This is the exact content of preserving sameness of length and difference of direction during composition.

The associated Hopf maps organize observation as well as composition. A normalized pair determines a relative description, while a whole sphere of states shares that description. What we read is the base; what it leaves unresolved is the fibre. The previous level's full state space reappears as the next level's fibre type. This is the recursive structure meant here by the fractality of the object: the comparison recurs inside a richer one.

AI formalism and analysis F16: One rule for the four Hopf maps

Status: established bundles and explicit compatibility.

For $A = \mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$, let $n = \dim_{\mathbb{R}} A$ and $|a|^2 + |b|^2 = 1$. Define

$$h_A(a, b) = (|a|^2 - |b|^2, 2a\bar{b}) \in \mathbb{R} \oplus A. \quad (25)$$

The norm identity gives

$$\|h_A(a, b)\|^2 = (|a|^2 - |b|^2)^2 + 4|a|^2|b|^2 = 1. \quad (26)$$

These are the division-algebra Hopf fibrations, with the sphere fibres shown in Table 2 ([Baez, 2002](#)).

AI formalism and analysis F16 (continued): One rule for the four Hopf maps

Their nontrivial sphere-over-sphere pattern is tied to the Hopf-invariant-one theorem (Adams, 1960).

Let $j : A \hookrightarrow B$ be one of the standard norm- and conjugation-preserving inclusions. With $J(a, b) = (j(a), j(b))$ and $K(t, x) = (t, j(x))$, substitution proves

$$h_B \circ J = K \circ h_A. \quad (27)$$

The larger projection retains the smaller one exactly. If $D(A) = A \oplus A$ is the next doubling for $A = \mathbb{R}, \mathbb{C}, \mathbb{H}$, then

$$h_{D(A)}^{-1}(1, 0) = \{(q, 0) : |q| = 1\} \cong S(A^2). \quad (28)$$

Thus the previous total space is exactly a distinguished next fibre. Other fibres require a chosen identification to carry that prior structure.

This is a finite structural recurrence. The next formula fails with sedenion multiplication: choose nonzero zero divisors $xy = 0$ with $|x| = |y| = 1/\sqrt{2}$, and set $a = x$, $b = \bar{y}$. Then the normalized input maps to $(0, 0)$ rather than to a unit sphere. The octonionic map already has total space S^{15} ; it remains valid because its formula uses octonionic multiplication. An octonionic fibre S^7 is not an associative phase group.

Algebra	Real dimension	Fibre	Total space	Base
$\mathbb{R}$	1	S^0	S^1	S^1
$\mathbb{C}$	2	S^1	S^3	S^2
$\mathbb{H}$	4	S^3	S^7	S^4
$\mathbb{O}$	8	S^7	S^{15}	S^8

Table 2: The same normalized comparison rule at four algebraic levels. Reading downward, each total-space type becomes the next fibre type. Reading across, projection retains a relative description while a fibre remains unresolved.

Binary is already inside this algebra. Label the quaternion units $1, i, j, k$ by 00, 01, 10, 11. Multiplication combines their bit labels by XOR and retains an additional sign. Since $ij = k$ and $ji = -k$, that sign records the difference between the two orders. This supplies a precise bridge between the cube and the sphere, without asking a reader to accept their resemblance as evidence.

AI formalism and analysis F17: Bit strings with an orientation sign

Status: established algebra and its interpretation.

The eight elements $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ form a group. Its commutator subgroup is $\{\pm 1\}$, since $[i, j] = -1$, while quotienting by these signs makes the generators commute. Hence

$$Q_8 / \{\pm 1\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2. \quad (29)$$

In the chosen labels the basis multiplication has the form $e_a e_b = \epsilon(a, b) e_{a \oplus b}$, with $\epsilon(a, b) \in \{\pm 1\}$. Related twisted group-algebra constructions extend this description to the octonions (Albuquerque and Majid, 1998).

Equation (29) forgets the sign information carried by quaternion multiplication. An ordinary binary representation can still encode the full signed units and their order. The thesis's claim that a binary reading is a projection is exact for this quotient; it is not a restriction on the expressive power of binary computers.

6.3 What the higher geometry gives the physical argument

The cup provides the intuition. A circular rim is visible from one direction, an ellipse from another, and a side view hides much of the interior. Moving around the cup changes the projection while offering more constraints on one object. In the algebraic model, the base and fibre make such retained and unresolved relationships explicit. A projection can be generated from a richer state even when the projected reading alone cannot recover that state.

Quaternions add three imaginary directions to a scalar component and compose spatial rotations without leaving their unit sphere. This makes them especially useful for the $3 + 1$ discussion. A physical spacetime description also has to account for its Lorentzian metric, where the time contribution and spatial contributions have opposite signs. The matrix representation below makes that extra structure explicit rather than leaving it inside the dimensional count.

AI formalism and analysis F18: Rotation, spacetime and observed quantum states

Status: established representations; dimensional interpretation.

For a unit quaternion $q = \cos(\theta/2) + \mathbf{n} \sin(\theta/2)$, conjugation $v \mapsto qv\bar{q}$ rotates the imaginary three-space by θ . The elements q and $-q$ give the same spatial rotation, yielding the double cover $SU(2) \rightarrow SO(3)$. Quaternion norm is Euclidean: $|q|^2 = w^2 + x^2 + y^2 + z^2$.

Minkowski coordinates instead form a Hermitian matrix

$$X = ctI + x\sigma_x + y\sigma_y + z\sigma_z, \quad \det X = c^2t^2 - x^2 - y^2 - z^2. \quad (30)$$

For $A \in SL(2, \mathbb{C})$, $X \mapsto AXA^\dagger$ preserves this determinant. Pauli matrices connect this representation with the complexified quaternion algebra. The signature is essential to its spacetime interpretation.

A normalized qubit $(a, b) \in \mathbb{C}^2$ lies on S^3 . Its global phase quotient is the Bloch sphere $\mathbb{CP}^1 \cong S^2$, the complex Hopf base. An outcome along a detector axis $\mathbf{n}$ has probability $(1 + \mathbf{r} \cdot \mathbf{n})/2$ by the Born rule, where $\mathbf{r}$ is the Bloch vector. Passing to the Bloch vector removes global phase; a subsequent measurement produces an outcome. They are distinct operations.

Poincaré–Bendixson restricts compact limit sets of suitable smooth autonomous planar flows: a nonempty omega-limit set without equilibria is a periodic orbit (Sideris, 2013). Three-dimensional flows can have richer recurrent dynamics. This supports the paper’s dimensional motivation, while neither it nor the quaternion coordinate count alone derives the dimension of spacetime.

Other connections belong here because they show what the same constructions already do. A periodic wave on a ring admits integer winding; bound-state spectra depend on boundary conditions and the operator whose eigenfunctions fit them. Two periodic motions at a rational frequency ratio return together, which can be heard as an interval and drawn as a closed Lissajous curve. Their lifts to suitable tori give a further topological description. These are concrete ways in which numbers, sounds and physical motions can be readings of one specified structure.

At the larger scale, division-algebra research constructs particle representations related to the Standard Model (Furey, 2016). Division algebras also explain the dimensions 3, 4, 6, 10 in certain classical supersymmetric constructions (Baez and Huerta, 2009). That dimension list has a different role from the quantum critical dimension of a string theory. The paper’s synthesis is that such recurring structures belong to one architecture of comparison and composition. Deriving a particular field content, action or particle spectrum still requires the corresponding physical construction. The common object is a research claim about those connections, and the Hopf recurrence states its mathematical core.

6.4 An exact account of a changed factor

Suppose a message is represented by an ordered sequence of unit rotations. Compose them, retain the successive products, and compare the record with a second sequence. What happens to a small change near the beginning? Unit multiplication carries it through without changing its size. A later position has the same sensitivity as an earlier one because both are compared through isometries.

This is useful in a way that a final verdict of “different” is not. We can measure the discrepancy, return to the local comparison, and ask whether a factor is absent or has changed position. The distinction has an exact algebraic form when a single elementary incident has been aligned.

AI formalism and analysis F19: Exact propagation and uniform sensitivity

Status: derivation from multiplicative norm.

Let $g_1, \dots, g_n$ be unit quaternions and $P_k = g_1 \cdots g_k$, with $P_0 = 1$. Replace only g_k by g'_k and hold all other factors fixed. Write $L = g_1 \cdots g_{k-1}$ and $R = g_{k+1} \cdots g_n$. Associativity gives

$$P_n - P'_n = L(g_k - g'_k)R, \quad |P_n - P'_n| = |g_k - g'_k|. \quad (31)$$

Both L and R are units, so the second equality follows from (23). Their real-linear action also preserves inner products, giving equality of the spherical distances before and after propagation. The differential $v \mapsto LvR$ is an isometry at every position. This is uniform sensitivity to one varied factor, with the others fixed.

Stored prefixes recover each factor exactly by $g_k = \bar{P}_{k-1}P_k$. For one nonzero substitution with common indexing, the predicate $P_j = P'_j$ is true before k and false thereafter. Binary search therefore locates k in $\lceil \log_2 n \rceil$ comparisons, given random access to exact stored prefixes. Constructing and storing them costs $O(n)$, and the claim counts comparisons, not the bit cost of arbitrary precision arithmetic. Multiple changes can make prefixes separate and rejoin.

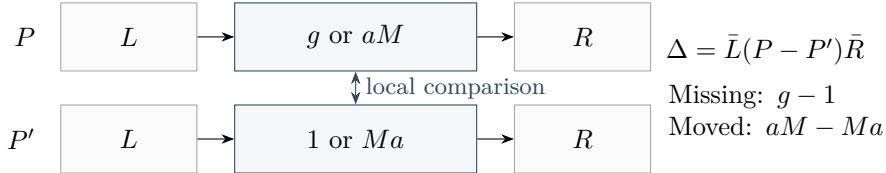


Figure 3: Remove the common context before classifying an elementary incident. An insertion leaves $g - 1$ in the local frame; a move leaves $aM - Ma$. The context changes the appearance of the discrepancy but preserves its norm.

An absent factor and a moved factor do different things. A nonidentity unit compared with the empty operation has a real component below the identity’s value. Moving a factor past a block compares two orders of the same product; their real parts agree, and the difference points in the perpendicular direction supplied by the cross product. This is a concrete realization of the distinction between presence and position.

AI formalism and analysis F20: The aligned elementary-incident discriminator

Status: exact result under the stated edit model.

Assume common unit contexts L, R have been identified. Define the additive local difference

$$\Delta = \bar{L}(P - P')\bar{R}. \quad (32)$$

AI formalism and analysis F20 (continued): The aligned elementary-incident discriminator

For insertion of a nonidentity unit g , $P = LgR$ and $P' = LR$, hence

$$\Delta = g - 1, \quad \text{Re } \Delta = \text{Re } g - 1 < 0. \quad (33)$$

A deletion reverses the sign. For a move past a block with product M , $P = LaMR$ and $P' = LMaR$, hence

$$\Delta = aM - Ma = 2 \text{Im } a \times \text{Im } M, \quad \text{Re } \Delta = 0. \quad (34)$$

The real-part assertion follows from (22); the vector part is perpendicular to both inputs. Thus a nonzero local discrepancy distinguishes these two incident classes exactly. For example, removing i gives $i - 1$ with real part -1 , while interchanging i, j gives $ij - ji = 2k$ with real part zero.

A commuting move gives $\Delta = 0$ and is invisible to the product. An identity factor is likewise invisible as an insertion. For several edits, or an unaligned window, the scalar test alone does not classify the history. Every finite sequence edit can be decomposed into elementary insertions, deletions and moves, but selecting that decomposition needs record alignment; it does not follow from one product. Keeping the stored path and distinguishable records makes that further inspection possible.

The ratio $P'^{-1}P$ measures the relative rotation and has a generic scalar part for either incident. Classification uses the aligned additive difference (32), rather than the scalar/vector display of that ratio.

6.5 The same calculation on unitary groups

The transport proof depends on multiplication being an isometry, so it extends beyond quaternion notation. In a sequence of unitary matrices, a changed gate is carried through the common surrounding gates at exactly its original norm. Trace then takes the role played by the scalar part: it ignores a cyclic exchange of factors but detects a nonidentity unitary against the identity.

This gives a possible experimental route. Known gate sequences can be compared after deliberate omissions and elementary reordering. The protocol has to isolate the incident and estimate the local quantity accurately enough to resolve it. The algebra predicts what to measure; laboratory repetitions establish the uncertainty of that measurement.

AI formalism and analysis F21: Unitary matrices and the higher-algebra scope

Status: exact extension and finite-precision analysis.

For $L, R, G, G' \in \text{U}(d)$ and either the operator or Frobenius norm,

$$\|L(G - G')R\| = \|G - G'\|. \quad (35)$$

For a move, $\text{tr}(AM - MA) = 0$. For an insertion,

$$\text{Re tr}(G - I) = \sum_{j=1}^d (\cos \theta_j - 1) < 0 \quad (G \neq I), \quad (36)$$

where $e^{i\theta_j}$ are the eigenvalues of G . Equivalently, $\|G - I\|_F^2 = 2d - 2 \text{Re tr } G$. These prove the same aligned discriminator. Trace estimation has a quantum computational realization in the one-clean-qubit model (Knill and Laflamme, 1998); it is an estimate, with sampling error.

Several factor errors obey $\|\prod G_j - \prod G'_j\| \leq \sum_j \|G_j - G'_j\|$ by telescoping. There is no corresponding positive lower bound: errors can cancel. Near the identity, $G = e^{\epsilon X}$ with $X^\dagger = -X$ gives

$$\text{Re tr}(G - I) = -\frac{1}{2}\epsilon^2 \|X\|_F^2 + O(\epsilon^3). \quad (37)$$

AI formalism and analysis F21 (continued): Unitary matrices and the higher-algebra scope

The presence residual is first order in norm but second order in its scalar test, a material consideration for finite precision.

For octonions with a fixed multiplication tree, varying one leaf is followed by a sequence of left or right unit multiplications, each an isometry. Single-leaf propagation remains exact. Rebracketing changes the operation, and no associative $L(\cdot)R$ factorization is available in general. An octonionic multi-edit classification requires an additional protocol; the quaternion proof is not silently reused for it.

The resulting noise account is deterministic about an algebraic comparison: given the two represented sequences and the aligned incident, the discrepancy has a definite size and form. This complements capacity, which asks what can be inferred through a channel with specified uncertainty. A physical source of error may still be unobserved, stochastic or quantum; its representation and measurement determine what the diagnostic can recover.

6.6 From a discrepancy to a field

Carry a reference around a closed path. On returning, it may be rotated from its starting orientation even though its length is unchanged. The loop has recorded something about the transport along the path. Physics calls this holonomy, and local failures of return are described by curvature. Here the closure comparison has become a measurement of an interaction.

The same sequence operation appears on a lattice: attach a transport to each link and multiply around a loop. A change of reference at the intermediate vertices cancels out; a trace gives a quantity independent of the starting frame. Wilson used such loop quantities to formulate lattice gauge theory (Wilson, 1974). Berry's phase is another physical realization of a return that carries geometric information (Berry, 1984).

AI formalism and analysis F22: Holonomy as a closure measurement

Status: established geometry and physical use.

For unitary transports U_{ab} from frame b to frame a , a local change of frame G_a acts as $U_{ab} \mapsto G_a U_{ab} G_b^{-1}$. For a loop based at a , its ordered product W transforms as

$$W \mapsto G_a W G_a^{-1}, \quad \text{tr } W \mapsto \text{tr } W. \quad (38)$$

The intermediate frame factors cancel explicitly. For a smooth connection, transport around an oriented coordinate rectangle shrinking at fixed aspect ratio, with area $\mathcal{A}$, has expansion $W = I + F_{\mu\nu} \mathcal{A} + o(\mathcal{A})$, up to the chosen sign convention. The curvature is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]. \quad (39)$$

The commutator is the contribution from order-dependent composition. For a compact Lie group with a bi-invariant metric and orthonormal tangent vectors X, Y , sectional curvature satisfies $K(X, Y) = \frac{1}{4} \|[X, Y]\|^2$ (Milnor, 1976).

These equations make a known transport discrepancy interpretable as a field measurement. Recovering an unknown field from finite noisy observations remains an inverse problem. A flat connection has trivial sufficiently small contractible-loop holonomy, while noncontractible loops can retain global information. Local closure and global topology therefore have separate roles.

6.7 How much must a message retain?

No representation travels with infinite experimental precision. If two observers need only agree to within one degree, they can use a finite angular code. If they need the winding count as well, it must also be transmitted or reconstructed from a sampled history. If the reference drifts, some of the precision budget must be spent on recalibration. Making these costs explicit connects the geometric unit back to the channel that carries it.

AI formalism and analysis F23: Finite resolution and a complete error budget

Status: elementary coding bound.

An angular code with m equally spaced representatives has worst-case nearest-point error π/m . For tolerance $0 < \varepsilon < \pi$, covering the circle requires at least

$$m \geq \left\lceil \frac{\pi}{\varepsilon} \right\rceil, \quad b \geq \lceil \log_2 m \rceil \quad (40)$$

fixed-length binary digits. Indeed, one representative covers an arc of length at most 2ε and the full length is 2π ; equal spacing attains the bound. Variable-length coding can exploit a distribution of readings, returning to Shannon's entropy.

If angular measurement, quantization, channel reconstruction and reference alignment have worst-case errors $\varepsilon_m, \varepsilon_q, \varepsilon_c, \varepsilon_r$, then the circular triangle inequality gives

$$\varepsilon_{\text{total}} \leq \min\{\pi, \varepsilon_m + \varepsilon_q + \varepsilon_c + \varepsilon_r\}. \quad (41)$$

A guaranteed winding count additionally requires a lift or a history with adequately bounded intersample motion; endpoint agreement cannot supply it. The bound makes the phrase "carrying the reference" operational: reference precision has a measurable cost in the protocol.

We can now operate the proposed unit through a complete chain. The circle defines a reference and a residual; a reading makes a finite distinction; composition preserves specified relations; a local comparison diagnoses particular changes; and a channel carries the description to an agreed precision. The next step is to say what a successful check tells us about meaning and action.

7 Meaning, action and a shared language

If I ask for the lighter and you hand it to me, your action supplies another reading of the request. I can compare the result with what I meant. Level C closes a practical loop through Level B: an interpretation leads to an action, the action produces an observation, and the observation lets us revise our understanding of whether the meaning arrived.

The reference has to be useful at both ends. Sending the name of a unit helps only if the receiver can realize that unit; sending a diagram helps only if its orientation and scale can be recovered. With a geometric bit we can include the operation, reference and return criterion in the message protocol. Their agreement becomes something the receiver can demonstrate. The construction gives a complete solution for the circular comparison task, and a specification for what richer meanings have to retain.

7.1 What an operational answer preserves

Suppose two descriptions differ in details that never affect the question being asked. A receiver can answer correctly without recovering every detail. If the question is which object to pick up, its exact temperature may be irrelevant. If the question is whether it is safe to touch,

temperature matters. Meaning therefore needs a declared task and enough context to distinguish the answers that the task makes different.

AI formalism and analysis F24: When a representation preserves a meaning

Status: necessary and sufficient recovery condition.

Let X be the admissible states, $q : X \rightarrow A$ the answer relevant to a task, and $E : X \rightarrow S$ the transmitted representation. A decoder $D : E(X) \rightarrow A$ satisfying $D(E(x)) = q(x)$ exists if and only if

$$E(x) = E(y) \implies q(x) = q(y). \quad (42)$$

Proof. Necessity follows by applying D to equal representations. For sufficiency, assign $D(s) = q(x)$ for any x with $E(x) = s$; the implication ensures that this value is independent of the representative. $\square$

For discrete random variables with the specified support, the probabilistic form is $H(q(X) | E(X)) = 0$, meaning exact recovery almost surely. If operations must remain usable, require Box F03 for those operations too. This is an exact Level B criterion for a declared task. It requires an interpretation of q and a procedure for constructing or accessing D ; mere equality of geometric summaries is insufficient when those summaries identify states with different answers.

Learning a relation often changes what a person can infer without further instruction. Someone who learns that an igniter is a lighter, and that the lighter is the object on the desk, can reach the object through the new word. Relational frame theory studies such learned relating through mutual entailment, combinatorial entailment and transformation of functions (Hayes et al., 2001; Association for Contextual Behavioral Science, n.d.). Coordination is one relational frame among others; comparison, opposition and perspective also change how a sign functions.

This is a useful bridge from the geometry to psychology because it suggests observable tests. Ask a person to reverse a learned relation, compose it with another or apply it under a changed reference. Their responses reveal which relationships became available. A successful test concerns those relationships under the specified context, rather than every association the person might have with the word.

AI formalism and analysis F25: Action, feedback and calibration

Status: operational model.

Let a be the intended task value, $\hat{a} = D(s)$ the receiver's interpretation, and $u = \pi(\hat{a}, c)$ an action chosen in context c . The environment produces an observation $y = O(u, c)$. An agreed target $y_*(a, c)$ and loss ℓ define the Level C check

$$L_C = \ell(O(\pi(D(s), c), c), y_*(a, c)). \quad (43)$$

Level B checks $D(s)$ against a ; Level C checks the resulting outcome. Feedback sends y back so that a reference or policy can be updated. Correct interpretation does not by itself guarantee a chosen action or a reliable actuator, but these failures can be distinguished when both readings are kept.

For a circular calibration example, let a frame offset be α and an estimate $\hat{\alpha}_t$. An ideal feedback reading supplies the signed residual $r_t = \text{Arg}(e^{i(\alpha - \hat{\alpha}_t)})$. With gain $0 < \eta \leq 1$, set

$$\hat{\alpha}_{t+1} = \hat{\alpha}_t + \eta r_t \pmod{2\pi}. \quad (44)$$

For an initial principal error strictly within $(-\pi, \pi)$ and noiseless feedback, the error contracts by $1 - \eta$ per step. This follows by substitution before any branch crossing. Repeated action can therefore calibrate a shared reference under stated observability conditions. Feedback with bias, drift or measurement error requires its own error budget.

7.2 A test that follows the argument

Give a sender and receiver differently oriented circular displays. On each trial the sender receives an instruction expressed in degrees, radians or a symbolic rotation. The receiver applies it on their own display. Retain the sent symbols, the interpreted angle, the final pointer position and the feedback response. We can then distinguish technical delivery, semantic recovery and successful action within one experiment.

The critical comparison is between a protocol that supplies an executable reference and one that relies on an assumed reference. Include identical-frame trials, controlled frame changes, errors of notation and deliberate channel corruption. Equalize the available communication budget, counting calibration data and feedback in that budget. A benefit must appear in the measured recovery and its cost, rather than follow from giving one condition extra information without accounting for it.

AI formalism and analysis F26: Evaluation and reproducibility

Status: proposed experiment; predictions of the model.

For trial j , record intended relative angle θ_j , decoded angle $\hat{\theta}_j$ and executed endpoint z_j . Define

$$L_B^{(j)} = |\text{Arg } e^{i(\hat{\theta}_j - \theta_j)}|, \quad L_C^{(j)} = d(z_j, e^{i\theta_j}), \quad (45)$$

after expressing all observations in the calibrated comparison frame. Record symbol error separately as Level A loss. Winding tasks add an integer loss.

The shared-reference protocol predicts invariance of L_B under a common rotation of its reference and reading, as proved in Box F09. Removing or corrupting that reference can destroy recovery; quantization and calibration predict the bounds in Box F23. For a machine version, insert known single-factor changes and aligned moves, checking Boxes F19–F21 against stored histories, including commuting moves and cancelling edits. These cases test the stated scope rather than only favourable examples.

Report distributions of angular error, failure rates, transmitted bits, calibration cost and latency. Human transfer to an untrained notation is a separate outcome. This manuscript gives mathematical predictions and numerical checks of its algebra; it reports no completed human-participant study.

7.3 What the larger thesis commits us to

The central proposal is that physics, mathematics and computation are different readings of an architecture of preserved relations. Physics tests it through interaction, mathematics expresses its constraints, and computation performs its operations. The circle and its compatible higher constructions give that proposal an explicit mathematical object. Extending the account to arbitrary meanings requires showing which observations and operations each meaning preserves, and how its representation makes those answers available.

Related empirical work offers places to investigate this extension. Learned representations can align across models and tasks; the Platonic representation hypothesis studies such convergence (Huh et al., 2024). Rotary position embeddings already use relative rotations in language-model computations (Su et al., 2021). Both make relational structure measurable. Neither establishes that a single circle or quaternion contains a complete representation of every meaning. The proposed research direction is to test what an explicit reference and compositional check add to these existing representations.

Biology makes the persistence question equally concrete. Experiments on planarian regeneration show that interventions in bioelectric signalling can alter later anatomical outcomes, including persistent changes in regenerative pattern (Durant et al., 2017). The question here is which relations keep a target form available through material change, and how a perturbation

changes the system's return. Such experiments provide observations against which a geometric account can be compared; a successful comparison must predict the measured outcomes, not merely assign them geometric names.

For computation, the immediate application is a protocol that stores enough history to compare ordered records, carries a reference for their interpretation, and exposes a residual the receiver can inspect. The Closure SDK is the implementation track of this programme. Its current implementation and performance require their own reproducible evaluation; this paper's algebraic proofs specify what that implementation should preserve. A product summary can be an efficient check while the retained record supplies the evidence needed to explain or resolve a disagreement.

7.4 Conclusion

We began with a lighter and the difficulty of carrying what we know about it to another person. The observer, the observation and the sign kept appearing together because each supplies something the other two need for a comparison. Shannon makes the choice countable once its alternatives are fixed; the present account asks how their identity can remain available when a description, viewpoint or operation changes.

The geometric bit operationalizes the identity principle through a referenced circle. Perpendicularity supplies a distinct direction, symmetry relates its positions, and Euler's formula composes the rotation that checks return. Discrete readings recover binary choices, while phase and winding retain different aspects of the continuous comparison. Composing the unit leads to joint state spaces and ordered products; the normed division algebras and their compatible Hopf maps state a precise recursive structure. Within that setting, exact error transport and aligned local comparison give concrete tests of what a composition preserved.

Meaning and action then have an operational form: specify the relation to be carried, make its reference usable, and compare what the receiver can recover and do. This settles the circular task and provides a testable specification for broader ones. The unifying thesis is information as invariance over change, with the same architecture read across disciplines. Its development proceeds by giving each proposed reading a construction, a consequence and a check that another observer can perform.

When I say A , the aim is that you reach A through your own reading, and that both of us have a way to check the journey. The paper offers a unit for making that requirement explicit, and a collaborative process in which the argument itself is subjected to it.

A Notation and the formal chain

Symbol	Meaning in this manuscript
X, R, Y	States, measurement contexts, and readings (F01).
E, D	Encoding and decoding maps, with the task and context specified (F02).
$T, \hat{T}$	A transformation and its represented operation (F03).
$H, I(X; Y)$	Shannon entropy and mutual information, measured in bits (F04–F06).
J, i	A quarter rotation on an oriented Euclidean plane (F07).
$\mathcal{G}$	The circular comparison unit, including its reference and metric (F08).
$r, z, c(r, z)$	Reference, reading, and relative phase (F09).
d, ε	Circular angular distance and closure residual, in radians (F08–F10).
S^k	The unit sphere of real dimension k , lying in $\mathbb{R}^{k+1}$.
h_A	The normalized-pair Hopf map over the algebra A (F16).
P_k	Ordered product through record k (F19).
L, R, Δ	Common product contexts and the aligned additive discrepancy (F20); here R is a product, not the context set of F01.
$W, F_{\mu\nu}$	Loop transport and connection curvature (F22).
q, L_B, L_C	Task answer and semantic/action losses (F24–F26).

The proof dependency is deliberately short. F01–F03 define the communication requirements. F04–F06 supply the quantitative background. F07 constructs the rotation used by F08–F10. F11–F13 state its physical and epistemological interpretation. F14–F18 develop the composition spaces; F19–F21 use the norm and alignment to prove the error results. F22 reads ordered transport as a field comparison, and F23 supplies finite precision. F24–F26 connect recovery to task meaning, action and an experiment.

B Source map and collaboration record

The manuscript follows the finished presentation’s seven chapters, recorded on 4 October 2026. The source snapshot and its hash are retained in the backup manifest. The following frame identifiers locate the intuitive sources without requiring a reviewer to infer them from an older draft.

Paper section	Presentation frames and source notes
1	answers, disciplines, relation, roles, fields ; the original Markdown argument on observation.
2	communicate, weaver, levels-ab, frege, univocity ; the degrees/radians example in the Markdown draft.
3	solved through information ; original Shannon and Weaver sources; the prior draft’s counting and binary geometry.
4	shape, alphabet, pi-e, discrete, checks, two-references, closure, reading ; <code>pending-paper-additions.md</code> , sections 1, 2, 8, 9.
5	law through s5-summary ; <code>PAPER_ANALYSIS.md</code> and its physical interpretation of the constants.
6	index-6 through compared ; <code>hopf-recursive-bridge.md</code> , <code>THE_OBJECT.md</code> , <code>pending-paper-additions.md</code> , section 10, and <code>PAPER_ANALYSIS.md</code> , sections 3, 7, 8.
7	levelc, life, humanities, machine-meaning, theory-of-theories, seed, coda ; the task-recovery and feedback arguments of the earlier full draft.

The external results are cited below. The author’s unpublished notes supply the synthesis, examples and proposed interpretations; their presence is not treated as independent confirmation. F19–F21 reorganize and refine the error derivations in those notes. The accompanying scripts check representative identities and boundary cases; proof remains in the displayed argument.

Walter is the human originator of the thesis and directs its public development. Astra wrote this edition from the presentation and source record, including the formal analysis and explicit scope decisions. Claude’s contribution at this stage is the earlier source analysis; a fresh review of this edition is the next step. The review ledger records agreement or disagreement against the F-identifiers, with proposed revisions and the author’s disposition. No result is labelled jointly verified merely because two AI systems appear in the contributor credit.

Data, code and availability

The source manuscript, vector figures, numerical checks, writing directives and review ledger accompany this draft. The presentation and release repository are at <https://github.com/faltz009/geometrical-theory-of-communication>. This edition contains no new human-subject data, and no claim of completed journal peer review. The work is shared under the repository’s CC BY-NC-SA 4.0 licence. Submission formatting and the outlet’s disclosure requirements will be applied when an outlet is selected.

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